

Infixation on the CV Tier: Evidence from Alabama Agent Agreement

Supplementary Handout, May 1 2026

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1 Relevant Rules

(1) Vocabulary Insertion Rules

1. $\text{AGR}_{[2, \text{SG}, \text{agent}]} \longleftrightarrow ch$
2. $\text{AGR}_{[2, \text{PL}, \text{agent}]} \longleftrightarrow hach$
3. $\text{AGR}_{[1, \text{PL}, \text{agent}]} \longleftrightarrow hili$ / two-mora final foot
4. $\text{AGR}_{[1, \text{PL}, \text{agent}]} \longleftrightarrow l$ / elsewhere
5. $\text{AGR}_{[1, \text{SG}, \text{agent}]} \longleftrightarrow a$ / _ NEG
6. $\text{AGR}_{[1, \text{SG}, \text{agent}]} \longleftrightarrow \emptyset$ / elsewhere
7. $\text{NEG} \longleftrightarrow k$
8. $\text{AUX} \longleftrightarrow o$

(2) Pivot Rules

- If right-most V is filled, then:
 - Insert the infix between the first $V_{i, [+FILLED]} C_{i+1, [+FILLED]}$ sequence from the right. Empty CV slots are ignored.
 - * Agent agreement marker is inserted to the left of $C_{i+1, [+FILLED]}$.
 - * Negation marker is inserted to the right of $V_{i, [+FILLED]}$.
 - If there is no instance of $V_{i, [+FILLED]} C_{i+1, [+FILLED]}$, place the agreement marker to the left of the left-most filled slot (i.e. prefixing iCV verbs).
- Else (i.e. right-most V is unfilled):
 - Insert the <C V> material at the end of the word (i.e. as a suffix).

(3) **Phonological Rules**

- Ungoverned V slots undergo *i*-insertion.
- *i*-insertion applies to monosyllabic roots.
- *i*-insertion applies to consonant-final stems.
- Complex segments (i.e. *ch*) must be in a licensed C.
- $V \rightarrow \emptyset / _ V$

2 Positive Agreement

TAM morphology is ignored, mostly for space reasons.

2.1 CVVCV

hinaata → *hinaa<s>ta*

(4) a. Underlying form *hinaata*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄
h	i	n	a			t	a

b. Insert material containing *ch* to the left of C₄

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	<C ₅	V ₅ >	C ₄	V ₄
h	i	n	a			ch		t	a

c. *ch* is ungoverned and unlicensed, spirantization to *s*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	<C ₅	V ₅ >	C ₄	V ₄
h	i	n	a			s		t	a

d. *a* spreads to V₃

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	<C ₅	V ₅ >	C ₄	V ₄
h	i	n	a			s		t	a

2.2 CVCCV

hokcho → *ho<chi>kcho*

- (5) a. Underlying form *hokcho*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃
h	o	k		ch	o

- b. Insert material containing *ch* before C₂

C ₁	V ₁	<C ₄	V ₄ >	C ₂	V ₂	C ₃	V ₃
h	o	ch		k		ch	o

- c. V₄ is ungoverned, insert *i*

C ₁	V ₁	<C ₄	V ₄ >	C ₂	V ₂	C ₃	V ₃
h	o	ch	i	k		ch	o

2.3 CVCV

hompani → *hompanchi*

- In CVCV verb, the final V in the timing tier is not associated with its melodic material. It links only when the agent agreement marker is not present.

- (6) a. Underlying form *hompani*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄
							⋮
h	o	m	a	p	a	n	i

- b. Insertion of agreement marker *hompanch*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄	C ₅	V ₅
							⋮		
h	o	m	a	p	a	n	i	ch	

- c. Insertion of *i* in word final position

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄	C ₅	V ₅
							⋮		
h	o	m	a	p	a	n	i	ch	i

2.4 iCV Verbs

ipa → *ilpa*

- (7) a. Underlying form *pa*

C ₁	V ₁	C ₂	V ₂
		p	a

- b. Insert *ch* material to the left of C₂

C ₁	V ₁	<C ₃	V ₃ >	C ₂	V ₂
		l		p	a

- c. *i* is inserted into V₁ (no monosyllabic stems)

C ₁	V ₁	<C ₃	V ₃ >	C ₂	V ₂
	i	l		p	a

2.5 -ka Verbs

- (8) Underlying form *-ka*

C ₁	V ₁	C ₂	V ₂
	∅	k	a

loska → *los<is>ka*

- (9) a. Underlying form
- loska*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄
l	o	s			∅	k	a

- b. Insert
- ch*
- material to the left of C
- ₄

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	<C ₅	V ₅ >	C ₄	V ₄
l	o	s			∅	ch		k	a

- c. Empty CV segment carrying ∅ deletes at some stage of the derivation

C ₁	V ₁	C ₂	V ₂	<C ₅	V ₅ >	C ₄	V ₄
l	o	s		ch		k	a

- d. V
- ₂
- is ungoverned, so we insert a
- i*
- to it

C ₁	V ₁	C ₂	V ₂	<C ₅	V ₅ >	C ₄	V ₄
l	o	s	i	ch		k	a

- e. Spirantization, since C
- ₃
- is neither governed nor licensed

C ₁	V ₁	C ₂	V ₂	<C ₅	V ₅ >	C ₄	V ₄
l	o	s	i	s		k	a

2.6 -li Verbs

bitli → *bitchi*

- (10) a. Underlying form
- bitli*

C ₁	V ₁	C ₂	V ₂
b	i	t	

- b. Insert after V
- ₂
- , since right-most V is unfilled

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃
b	i	t		ch	

- c.
- i*
- insertion at end of word

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃
b	i	t		ch	i

2.7 Remaining Verbal Stems

- Periphrastic: *ootoba-t-il-ka* “we dream”
- Causative: *aaba-chi-l-ka* “we teach”

★ This dervies the positive agreement paradigms!

3 Derivations in Strict CV: Negative Agreement

- The negation paradigm always suffixed with the auxiliary *o*, which triggers vowel deletion of the previous vowel:
 - e.g. *choopa* → *cho<chi-kii>pa* → *cho<chi-kii>pa + o* → *cho<chi-kii>po* → *cho<chi-kii>po+ti*
- The derivations here do not include the negative auxiliary *o* and TAM morphology, for space reasons.

3.1 CVCCV

hokcho → *hochikikcho*

(11) a. Underlying form *hokcho*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃
h	o	k		ch	o

b. Insert negation and agreement material

C ₁	V ₁	<C ₄	V ₄	C ₅	V ₅ >	C ₂	V ₂	C ₃	V ₃
h	o	ch		k		k		ch	o

c. All ungoverned slots filled at once

C ₁	V ₁	<C ₄	V ₄	C ₅	V ₅ >	C ₂	V ₂	C ₃	V ₃
h	o	ch	i	k	i	k		ch	o

- Note that this does suggest that the “ungoverned” slots all at once, to ensure that V₄ is filled

3.2 CVVCV

- (12) a. 1. Underlying form *choopa*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃
ch	o			p	a

- b. 2. Infixation of NEG, right of V₁

C ₁	V ₁	<C ₄	V ₄ >	C ₂	V ₂	C ₃	V ₃
ch	o	k				p	a

- c. 3. Infixation of 2sg, left of C₄

C ₁	V ₁	<C ₅	V ₅ >	<C ₄	V ₄ >	C ₂	V ₂	C ₃	V ₃
ch	o	ch		k				p	a

- d. 4. *i*-insertion: V₅ and V₄ are ungoverned

C ₁	V ₁	<C ₅	V ₅ >	<C ₄	V ₄ >	C ₂	V ₂	C ₃	V ₃
ch	o	ch	i	k	i			p	a

- e. 5. Remaining affixes & surface phonology: *cho<chi-kii>p-o-ti* “you didn’t buy’

3.3 CVCV

aʔi → *aʔchiko*

- (13) a. *aʔi*

C ₁	V ₁	C ₂	V ₂
			⋮
	a	ʔ	i

- b. Agent agreement affixation

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄
			⋮				
	a	ʔ	i	ch		k	

- c. *i*-insertion in ungoverned slot

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄
			⋮				
	a	ʔ	i	ch	i	k	

3.4 iCV

ila → *chiklo*

- (14) a. Underlying form *ila*

C ₁	V ₁	C ₂	V ₂
		l	a

- b. NEG and AG insertion

C ₁	V ₁	<C ₃	V ₃	C ₄	V ₄ >	C ₂	V ₂
		ch		k		l	a

- c. *i*-insertion into ungoverned slots

C ₁	V ₁	<C ₃	V ₃	C ₄	V ₄ >	C ₂	V ₂
		ch	i	k		l	a

3.5 -ka verbs

loska → *loschikko*

- (15) a. Underlying *loska*

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	C ₄	V ₄
l	o	s			∅	k	a

- b. <AG NEG> insertion

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	<C ₅	V ₅	C ₆	V ₆ >	C ₄	V ₄
l	o	s			∅	ch		k		k	a

c. *i*-insertion

C ₁	V ₁	C ₂	V ₂	C ₃	V ₃	<C ₅	V ₅	C ₆	V ₆ >	C ₄	V ₄
l	o	s			∅	ch	i	k		k	a

d. ∅ material is deleted

- Note that by comparing the positive and negative agreement derivations, the *i*-insertion for un-governed slots occurs twice:
 - It is evaluated *before* the ∅ material is deleted, shown in (15)-c
 - It is evaluated *after* the ∅ material is deleted, shown in (9)-c

3.6 Remaining Verbal Stems: *-li* and *-chi* Verbs

- The remaining verbs are *-li*, *-chi* verbs
- **Todo:** I need to still provide derivations of these, but these are similar because they seem to involve periphrastic conjugation. See table 5.
- **Todo:** The other issue is the relative ordering of AG and NEG in 1SG.
 - In 1SG for all verbal stems the order of morphemes is <a - k > *except in CVVCV and CVCCV, which as <k a(a)>*
 - *cho<k-aa>po-ti* ‘we didn’t buy’
 - *los<ak>ko-ti* ‘we didn’t buy’
- **Todo:** Note that in 1SG, there is a *t* between the verbal stem and agreement marker *a*¹<tak>ko
 - This could suggest that CVCV verbs are actually periphrastic, and the /t/ only surfaces when the following segment is a vowel

4 Lexically Specified Conjugation Classes

- A small set of verbs belong to lexically specified conjugation classes
 - *nochi* ‘to sleep’
 - *hiicha* ‘to see’
 - *haalo* ‘to hear’
- These are CVCV/CVVCV stems but take with the iCV conjugation, as illustrated in Table 1.

	SG	PL
1	nochi-li-ti	il -nochi-ti
2	is -nochi-ti	has -nochi-ti
3	nochi-ti	(ho)-nochi-ti

Table 1: *nochi* ‘to sleep’

- (16) a. *nochi*
- | | | | | | |
|----------------|----------------|----------------|----------------|----------------|----------------|
| C ₁ | V ₁ | C ₂ | V ₂ | C ₃ | V ₃ |
| | | | | ⋮ | ⋮ |
| | | n | o | ch | i |
- b. *il-nochi*
- | | | | | | | | |
|----------------|----------------|-----------------|------------------|----------------|----------------|----------------|----------------|
| C ₁ | V ₁ | <C ₃ | V ₃ > | C ₂ | V ₂ | C ₃ | V ₃ |
| | | | | | | ⋮ | ⋮ |
| | | l | | n | o | ch | i |
- c. *i*-insertion for “monosyllabic” stem
- | | | | | | | | |
|----------------|----------------|-----------------|------------------|----------------|----------------|----------------|----------------|
| C ₁ | V ₁ | <C ₃ | V ₃ > | C ₂ | V ₂ | C ₃ | V ₃ |
| | | | | | | ⋮ | ⋮ |
| | i | l | | n | o | ch | i |
- d. melodic material linked (at some stage)
- | | | | | | | | |
|----------------|----------------|-----------------|------------------|----------------|----------------|----------------|----------------|
| C ₁ | V ₁ | <C ₃ | V ₃ > | C ₂ | V ₂ | C ₃ | V ₃ |
| | | | | | | | |
| | i | l | | n | o | ch | i |

5 Remaining Issues

• CCC Clusters

- There are still some situations with CCC clusters: *bit<k>ko* “S/he didn’t dance”
- Compare this with the 3SG negation of *ka* verbs: *los<ik>ko* “s/he didn’t lie”
 - This suggests that there’s something about *li* verbs that stops the *i* insertion.

• Suppletive Allomorphy Unexpected in 1PL Negation

- Under the assumptions adopted here, suppletive allomorphy should already have been determined at this stage of the derivation, which would lead us to expect only the form in (17-b) instead.

- (17) a. bit<kil>ko-ti
“We didn’t dance”
- b. bit<hilik>ko-ti
“We didn’t dance”

• **Suppletive Roots**

- Many motion verbs in Alabama exhibit suppletive roots conditioned by number. For example, *ałła* ‘to go’ shows distinct stems in singular, dual, and plural contexts, as illustrated in table 2, and *wiika* / *aswa* / *iisa* ‘to exist’, shown in table 3

	SG	DU	PL
1	ałła- li -ti	anłiiya-ti	amaa- il -ka-ti
2	ał- chii -ya-ti	ała- hachii -ya-ti	amaa- s -ka-ti
3	ałła-ti	ałła- chi -ti	amaa-ka-ti

Table 2: *ałła* ‘to go’

	SG	DU	PL
1	wiika- li -ti	a- li -swa-ti	iis(a)-t- il -ka-ti
2	wii- his -ka-ti	a- hachi -swa-ti	iis(a)t- as -ka-ti
3	wiika-ti	aswa-ti	iisa-ti

Table 3: *wiika/aswa/iisa* ‘to exist’

Syll. Structure	Infix Placement	Example Word	1PL Output	2SG Output	2PL Output
i.CV	<iX>CV	<i>i.pa</i>	<i>il-pa</i>	<i>is-pa</i>	<i>has-pa</i>
-VVka	-VV<(h)iX>ka	<i>afaa.ka</i>	<i>afaa<(h)il>ka</i>	<i>afaa<(h)is>ka</i>	<i>afaa<has>ka</i>
-VCka	-VC<iX>ka	<i>los.ka</i>	<i>los<il>ka</i>	<i>los<is>ka</i>	<i>los<has>ka</i>
Periphrastic	-t-X-ka	<i>ootoba</i>	<i>ootoba-t-il-ka</i>	<i>ootoba-t-is-ka</i>	<i>ootoba-t-as-ka</i>
CVV.CV	CVV<X>CV	<i>hi.naa.ta</i>	<i>hinaa<l>ta</i>	<i>hinaa<s>ta</i>	<i>hinaa<(h)as>ta</i>
-chi	-chi<X>ka	<i>aaba.chi</i>	<i>abaa-chi<l>ka</i>	<i>aaba-chi<s>ka</i>	<i>abaa-ch<as>ka</i>
CV.CV	CVC<X>	<i>hom.pa.ni</i>	<i>hompan-hili</i>	<i>hompan-chi</i>	<i>hompan-hachi</i>
-li	-Xi	<i>bit.li</i>	<i>bit-hili</i>	<i>bit-chi</i>	<i>bit-hachi</i>
-li	-Xi	<i>soo.li</i>	<i>soo-hili</i>	<i>soo-chi</i>	<i>soo-hachi</i>
CVC.CV	CV<Xi>CCV	<i>hokcho</i>	<i>hokcho</i>	<i>ho<chi>kcho</i>	<i>ho<hachi>kcho</i>

Table 4: Positive Agent Infix placement, organized by agent allomorph

Syll. Structure	1SG	1PL	2SG	2PL	3SG
<i>i.la</i>	< <i>ak</i> >lo	< <i>kil</i> >lo	< <i>chik</i> >lo	< <i>hachik</i> >lo	< <i>ik</i> >lo
<i>afaa.ka</i>	afaa<??>ko				
<i>los.ka</i>	los< <i>ak</i> >ko	los< <i>kil</i> >ko	los< <i>chik</i> >ko	los< <i>hachik</i> >ko	los< <i>ik</i> >ko
<i>choopa</i>	cho< <i>k-aa</i> >po	cho< <i>ki-lii</i> >po	cho< <i>chi-kii</i> >po	cho< <i>hachi-kii</i> >po	cho< <i>kii</i> >po
<i>aaba.chi</i>	abaa-ch< <i>ak</i> >ko	abaa-chi< <i>kil</i> >ko	abaa-chi< <i>chik</i> >ko	abaa-chi< <i>hachik</i> >ko	abaa-chi< <i>k</i> >ko
<i>ałi</i>	ał< <i>tak</i> >o	ał< <i>kil</i> >o	ał< <i>chik</i> >o	ał< <i>hachik</i> >o	ał< <i>k</i> >o
<i>bit.li</i>	bit< <i>tak</i> >ko	bit< <i>kil</i> >ko bit< <i>hilik</i> >ko	bit< <i>chik</i> >ko	bit< <i>hachik</i> >ko bit< <i>hask</i> >ko	bit< <i>k</i> >ko
<i>soo.li</i>	soo< <i>tak</i> >ko	soo< <i>kil</i> >ko bit< <i>hilik</i> >ko	soo< <i>chik</i> >ko	soo< <i>hachik</i> >ko	soo< <i>k</i> >ko
<i>hok.cho</i>	ho< <i>ka</i> >kcho	ho< <i>kili</i> >kcho	ho< <i>chi-ki</i> >kcho	ho< <i>hachi-ki</i> >kcho	ho< <i>ki</i> >kcho

Table 5: Negative Stem placement patterns across syllable structures

Syll. Structure	Infix Placement	Example Word	Output
i.CV	<iX>CV	<i>i.pa</i>	< <i>il</i> > <i>pa</i>
-VVka	-VV<(h)iX>ka	<i>afaa.ka</i>	??
-VCka	-VC<iX>ka	<i>los.ka</i>	???
Periphrastic	-t-X-ka	<i>ootoba</i>	??
CVV.CV	CVV<X>CV	<i>choopa</i>	<i>choo</i> < <i>l</i> > <i>pa</i>
-chi	-chi<X>ka	<i>aaba.chi</i>	<i>abaachi-t</i> < <i>il</i> > <i>ka</i>
CV.CV	CVC<X>	<i>ta!a</i>	<i>ta!</i> -< <i>ka</i> >
-li	-Xi	<i>batli</i>	<i>bat</i> -< <i>ka</i> >
-li	-Xi	<i>albitiili</i>	<i>albitii</i> -< <i>ka</i> >
CVC.CV	CV<Xi>CCV	<i>hosso</i>	<i>ho</i> < <i>li</i> > <i>sso</i>

Table 6: Middle Voice morphology placement patterns across syllable structures